THE ISOMORPHISM SPACE OF TWO OBJECTS OF AN ALGEBRAIC STACK IS LOCALLY OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The description of the isomorphism sheaf as a fibre product of the diagonal and the sheaf of isomorphisms of the base. Used inside the counted proof (source lines 286-296). Exhibits the isomorphism space as a pullback of the diagonal.
- The base change permanence and the description of the diagonal through a presentation. Invoked inside the counted proof (source lines 297-305). Transfers local finite type from the diagonal to the isomorphism space.

Source reference. Original reference algebraic-lemma-stack-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 287–287. Navigation pointer only: the counted text refers to the Stacks item algebraic-lemma-stack-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-descent-lemma-descending-property-locally-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 292–292. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-descending-property-locally-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-groupoids-lemma-quotient-stack-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

297–297. Navigation pointer only: the counted text refers to the Stacks item spaces-groupoids-lemma-quotient-stack-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-finite-presentation-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 305–305. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-finite-presentation-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-permanence-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 307–307. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-permanence-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-smooth-locally-finite-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 304–304. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-smooth-locally-finite-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let T be a scheme and let x, y be objects of the fibre category of $\mathcal{X}$ over T. Then the morphism $Isom_{\mathcal{X}}(x, y) \to T$ is locally of finite type.*

Proof. By Algebraic Stacks, Lemma algebraic-lemma-stack-presentation we may assume that $\mathcal{X} = [U/R]$ for some smooth groupoid in algebraic spaces. By Descent on Spaces, Lemma spaces-descent-lemma-descending-property-locally-finite-type it suffices to check the property fppf locally on T. Thus we may assume that x, y come from morphisms $x', y' : T \to U$. By Groupoids in Spaces, Lemma spaces-groupoids-lemma-quotient-stack-morphisms we see that in this case $Isom_{\mathcal{X}}(x, y) = T \times_{(y', x'), U \times_S U} R$. Hence it suffices to prove that $R \to U \times_S U$ is locally of finite type. This follows from the fact that the composition $s : R \to U \times_S U \to U$ is smooth (hence locally of finite type, see Morphisms of Spaces, Lemmas spaces-morphisms-lemma-smooth-locally-finite-presentation and spaces-morphisms-lemma-finite-presentation-finite-type) and Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-finite-type. □